Iniciación en la Investigación (Semester 1 2025)

Lecture 4: Preparations for LSST

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May 30, 2025

Motivation

Knowing specifics of a survey as far as possible in advance helps developing software/ algorithms so they are ready once the survey begins.

Simulations of LSST data assist with this process.

Cadence Notes

The Survey Cadence Optimization Committee (SCOC) makes recommendations for early observations (e.g., prioritization of sky area coverage) with early science goals in mind. The acquisition of commissioning data is also driven by the need to verify and validate the requirements on all Rubin Observatory systems and meet the construction completeness criteria.

The planned surveys for system optimization and science validation include pilot multi-band surveys that are shallow but a wide-area; reach 1-2 year depths over 100 to 1,000 square degrees, potentially including a region with dense temporal sampling; reach 10-20 year depths in a small area, such as a deep drilling field.

Cadence Notes

LSST Cadence Notes Solicitation (December 2020):

The scientific community was participating in the process of survey design by evaluating the simulated cadences:

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Only cadence given, but no simulated photometry.

Cadence Notes

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The cadence simulations explored offered variations on the survey strategy. Those different survey strategies are fine-tuned variations from a **baseline strategy** in an attempt to further enhance the science outcome from LSST.

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For those, what was varied were some of the fundamental survey parameters*, including:

- Survey footprint and distribution of visits
- Exposure time per visit
- Allocation of observing time per band - the distribution of visits between filters
- Time sampling (cadence) and dithers - on timescales from nightly to monthly to yearly

* more on this: `https://docushare.lsstcorp.org/docushare/dsweb/Get/Version-69049/LSSTCadenceNotesSolicitationV2.pdf`

Cadence Notes

some cadence notes:

Classical variable stars in different Galactic environments: pulsation behaviour recovery.

Ilaria Musella*, Marcella Di Criscienzo, Vittorio Braga, Silvio Leccia, Massimo Dall'Ora, Giuliana Fiorentino, Marcella Marconi, Roberto Molinaro, Vincenzo Ripepi, Giuseppe Bono, Michele Trabucchi, Leo Girardi, Giulia De Somma
with the support of the LSST Transient and Variable Stars (TVS) and Stars, Milky Way & Local Volume (SMWLV) Collaborations

April 15, 2021

Abstract

In this Cadence note, we present recommendations for Vera C. Rubin (Ivezić et al., 2019) observations to maximize the expected outcome for our favorite science case. In particular, we analyze the performance of different LSST OpSim strategies to recover accurate periods, mean magnitudes and amplitudes, together with a well sampled light curve in different stellar system distributed in distance for different types of pulsating stars characterized by different light curve shapes and ranges of period, absolute mean magnitude and amplitude (Type ab and c RR Lyrae, Cepheids, Long Period Variables, δ Scuti). To this aim we construct a metric that, starting from a theoretical model or an empirical template for a variable star, generates a light curve based on the LSST OpSim cadence and analyzes the goodness of the recovered period, mean magnitude and amplitude as a function of the number of years of observations. An integral part of this metric is the multi-filter Gatspy package that allows us to obtain a periodogram based on all the bands and to maximize the expected outcome.

Give Me a Few Hours: Missing Timescales in Rubin Cadence Simulations

Eric C. Bellm (UW)*, Colin Burke (Illinois), Michael Coughlin (UMn), Igor Andreoni (Caltech), Claudia M. Raiteri (INAF), and R. Bonito (INAF–OAPa),
with support of the LSST Transient and Variable Stars Collaboration

Abstract

In this Cadence Note, we analyze the distribution of visit separations for a range of Rubin Observatory cadence simulations. The limiting temporal resolution of a time-domain survey in detecting transient behavior is set by the time between observations of the same sky area. **Current simulations are strongly peaked at the 22 minute visit pair separation and provide effectively no constraint on temporal evolution within the night.** This choice will necessarily prevent Rubin from discovering a wide range of astrophysical phenomena in time to trigger rapid followup. We present a science-agnostic metric to supplement detailed simulations of fast-evolving transients and variables.

1 Science Case

As a supplement to other Cadence Notes which treat specific classes of fast transients (e.g., Andreoni et al.) and accreting sources (e.g., Raiteri et al., Bonito & Venuti et al.), we present a source-agnostic analysis of the time gaps present in current cadence simulations. While detailed metrics simulating specific object classes are important in determining the science impacts of specific cadence choices, they require extensive development by domain experts and may not span the discovery space. Additionally it is challenging to weight specific object classes against one another. We suggest here the utility of higher-level metrics in conjunction with domain knowledge.

The (logarithmic) time separation of visits to a given sky area encapsulates the most

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a list of all cadence notes:
`https://www.lsst.org/content/survey-cadence-notes-2021`

Data Previews

A series of Data Previews (DP) based on simulated and commissioning data are planned to support the community as they develop their LSST analysis software and workflows, and to enable high-impact science as soon as possible.

Data Previews serve as an early integration test of the Legacy Survey of Space and Time (LSST) Science Pipelines and the Rubin Science Platform (RSP).

They enable a limited number of scientists and students to begin early preparations for science with simulated LSST-like data sets.

Data Previews

Data Preview 0 (DP0) is the first of three data previews during the period leading up to the start of Rubin Observatory Operations.

DP0 users: The scientists and students who have Rubin Science Platform accounts and access to the DP0 data sets.

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It is available at the Rubin Science Platform from here: `https://dp0.lsst.io/index.html`

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How are the Data Previews created?

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Data Preview 2 (DP2) will be a full reprocessing of all science-grade commissioning data, and will include processed visit images, coadded images, difference image analysis, and associated catalogs. It is expected to be based on data from the LSST Science Camera, and to become available by May 2026.

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Data Release 1 (DR1) will be the first half-year of data from survey operations, and will include all types of data products. As processing is anticipated to take 6 months, DR1 is expected by Jan 2027.

Data Preview 0

DP0 comes in different versions:

DP0.1:
DP0.1 was retired on May 2 2024 and is no longer available, having been replaced by DP0.2 in June 2022. Please visit `dp0-2.lsst.io` for more information.

DP0.2: For simulated Galactic and extragalactic data products, see the documentation for DP0.2. DP0.2 also contains simulated light curves.

DP0.3:
The DP0.3 simulation only contains catalog data products for Solar System objects.

Data Preview 0

DP0 is based on the the simulated, Legacy Survey of Space and Time (LSST)-like images generated by the Dark Energy Science Collaboration (DESC) for their Data Challenge 2 (DC2; arXiv:2010.05926). DP0 uses the 300 deg^2 of DC2 images that were simulated for five years of the LSST's wide-fast-deep component (WFD; also called the main survey) with a baseline cadence.

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Simulated Objects and Images: The DC2's WFD simulated images include galaxies (with large-scale structure), Type Ia supernovae, and stars (10% of which are variable). DP0.1 does not include AGN, strong lenses, solar system objects, non-Ia extragalactic transients, or diffuse features (e.g., tidal streams, intracluster light). The DESC simulated the DC2 images using the imSim package.

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Image Processing: The DESC processed the simulated DC2 images with Version 19 of the LSST Science Pipelines. DP0.1 makes the DESC's DC2 images and catalogs available to RSP users. For DP0.2, the Rubin Data Production team reprocessed the same images with the most up-to-date version of the LSST Science Pipelines.

Data Preview 0

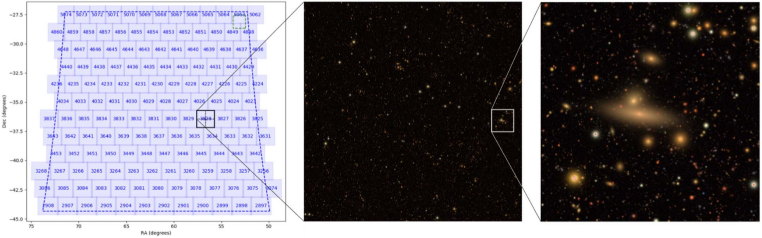

Figure 15 from The LSST DESC DC2 Simulated Sky Survey, showing the simulated wide-fast-deep (WFD) region divided into tracts. The center image is one tract quadrant, and the right image one hundredth the area of the tract quadrant. Patches are larger than the right image, as described in the DESC paper: *each tract is composed of 7 × 7 patches, and each patch is 4,100 × 4,100 pixels with a pixel scale of 0.2 arcsec*. Source: Abolfathi et al. (2021)

Data Preview 0

There was some work done so far with DP0 data.

As an example the following papers:

Data Preview 0

Predictions for the Detectability of Milky Way Satellite Galaxies and Outer-Halo Star Clusters with the Vera C. Rubin Observatory

Kabelo Tsiane†,[1,2] Sidney Mau†,[3,4] Alex Drlica-Wagner†,[5,2,6,7] Jeffrey L. Carlin,[8] Peter S. Ferguson,[9] Keith Bechtol,[10] Ethan O. Nadler,[11] Annika H. G. Peter,[12,13,14] Yao-Yuan Mao,[15] and Adam J. Thornton[16]
(LSST DESC Collaboration)

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[2] *Department of Astronomy and Astrophysics, University of Chicago, Chicago, IL 60637, USA*
[3] *Department of Physics, Stanford University, 382 Via Pueblo Mall, Stanford, CA 94305, USA*
[4] *Kavli Institute for Particle Astrophysics & Cosmology, P.O. Box 2450, Stanford University, Stanford, CA 94305, USA*
[5] *Fermi National Accelerator Laboratory, P.O. Box 500, Batavia, IL 60510, USA*
[6] *Kavli Institute for Cosmological Physics, University of Chicago, Chicago, IL 60637, USA*
[7] *NSF-Simons AI Institute for the Sky (SkAI), 172 E. Chestnut St., Chicago, IL 60611, USA*
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[9] *DiRAC Institute, Department of Astronomy, University of Washington, 3910 15th Ave NE, Seattle, WA, 98195, USA*
[10] *Physics Department, 2320 Chamberlin Hall, University of Wisconsin-Madison, 1150 University Avenue Madison, WI 53706-1390*
[11] *Department of Astronomy & Astrophysics, University of California, San Diego, La Jolla, CA 92093, USA*
[12] *The Ohio State University, Department of Physics, Columbus, OH 43210, USA*
[13] *The Ohio State University, Department of Astronomy, Columbus, OH 43210, USA*
[14] *Center for Cosmology and Astroparticle Physics, 191 West Woodruff Avenue, Columbus, OH 43210, USA*
[15] *Department of Physics and Astronomy, University of Utah, Salt Lake City, UT 84112, USA*
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Version April 24, 2025

ABSTRACT

We predict the sensitivity of the Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST) to faint, resolved Milky Way satellite galaxies and outer-halo star clusters. We characterize the expected sensitivity using simulated LSST data from the LSST Dark Energy Science Collaboration (DESC) Data Challenge 2 (DC2) accessed and analyzed with the Rubin Science Platform as part of the Rubin Early Science Program. We simulate resolved stellar populations of Milky Way satellite galaxies and outer-halo star clusters over a wide range of

Data Preview 0

Tsiane et al. (2025) predict the sensitivity of LSST regarding faint, resolved Milky Way satellite galaxies and outer-halo star clusters.

They characterize the expected sensitivity using simulated LSST data from the LSST Dark Energy Science Collaboration (DESC) Data Challenge 2 (DC2) as part of the Rubin Early Science Program.

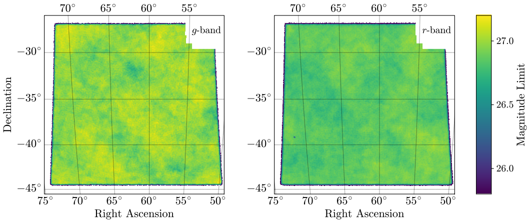

Fig. 1: The simulated LSST DESC DC2 footprint covers ~ 300 deg^2 of the high Galactic latitude sky. Maps of the S/N $= 5$ mag limit for point-like sources in the g and r band show the uniformity and depth of the WFD portion of DC2. The median S/N $= 5$ mag limits for point-like sources are $g = 27.0$ mag and $r = 26.8$ mag.

Data Preview 0

Tsiane et al. (2025) simulate resolved stellar populations of Milky Way satellite galaxies and outer-halo star clusters over a wide range of sizes, luminosities, and heliocentric distances, which are broadly consistent with expectations for the Milky Way satellite system. Simulated stars are into the DC2 catalog, having realistic photometric uncertainties and star/galaxy separation derived from the DC2 data itself.

The probability that each simulated system would be detected by LSST is assessed using a conventional isochrone matched-filter technique.

Conclusions: When assuming perfect star/galaxy classification and a model for the galaxy-halo connection fit to current data, they predict that 89 ± 20 Milky Way satellite galaxies will be detectable with a simple matched-filter algorithm applied to the LSST wide-fast-deep data set.

Data Preview 0

A&A, 686, A11 (2024)
https://doi.org/10.1051/0004-6361/202349012

Astronomy & Astrophysics

Recovered supernova Ia rate from simulated LSST images

V. Petrecca[1,2], M. T. Botticella[1], E. Cappellaro[3], L. Greggio[3], B. O. Sánchez[4], A. Möller[5], M. Sako[6], M. L. Graham[7], M. Paolillo[2,1,8], F. Bianco[9,10], and the LSST Dark Energy Science Collaboration

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ABSTRACT

Aims. The *Vera C. Rubin* Observatory's Legacy Survey of Space and Time (LSST) will revolutionize time-domain astronomy by detecting millions of different transients. In particular, it is expected to increase the number of known type Ia supernovae (SN Ia) by a factor of 100 compared to existing samples up to redshift ~1.2. Such a high number of events will dramatically reduce statistical uncertainties in the analysis of the properties and rates of these objects. However, the impact of all other sources of uncertainty on the

Data Preview 0

LSST is expected to increase the number of known type Ia supernovae by a factor of 100 compared to existing samples up to redshift of $\sim$1.2. Such a high number of events will dramatically reduce statistical uncertainties in the analysis of the properties and rates of these objects.

However, the uncertainty on themeasurement of the SN Ia rate must still be evaluated. The comprehension and reduction of such uncertainties will be fundamental both for cosmology and stellar evolution studies, as measuring the SN Ia rate can put constraints on the evolutionary scenarios of different SN Ia progenitors.

Methods: Petrecca et al. (2024) use simulated data from the Dark Energy Science Collaboration (DESC) Data Challenge 2 (DC2) and LSST Data Preview 0 to measure the SN Ia rate on a 15 deg^2 region of the WFD area. They select a sample of SN candidates detected in difference images, associated them to the host galaxy, and retrieved their photometric redshifts.
They then tested different light-curve classification methods, with and without redshift priors (albeit ignoring contamination from other transients, as DC2 contains only SN Ia).

Data Preview 0

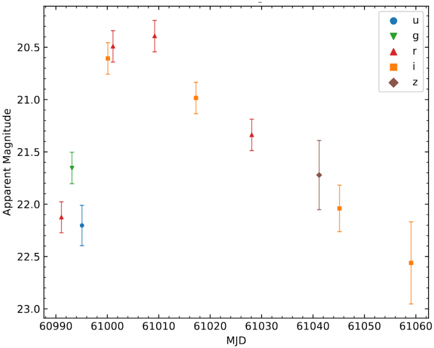

Fig. 2. Example multiband light curve of a SN Ia with redshift $z = 0.16$. Different shapes and colors refer to different filters, as reported in the legend.

Data Preview 0

Preparations for LSST

Motivation

Survey strategy

Data Previews

Further Resources

Results: Petrecca et al. (2024) find that 17% of the SN Ia of the simulated sample are missed. As 10% of the mismatch is due to the uncertainty on the photometric redshift alone (which affects classification when used as a prior), they conclude that this parameter is the major source of uncertainty.

They discuss possible reduction of the errors in the measurement of the SN Ia rate, including synergies with other surveys.

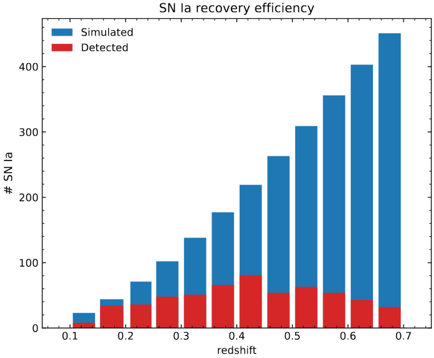

Fig. 9. Numbers of SN Ia simulated (blue) and detected (red, the SN Ia sample) in the redshift bins used for the measurement of the rate.

Early Science Program

"Early science" is defined as any science enabled by Rubin for its community during the commissioning period, the data previews, the start of alert production, and the first year of survey operations, up to and including Data Release 1 (DR1).

As the final stage of the Rubin construction project, commissioning began in 2024 with the Commissioning Camera, and proceeds in 2025 with the LSST Science Camera.

Early Science Program

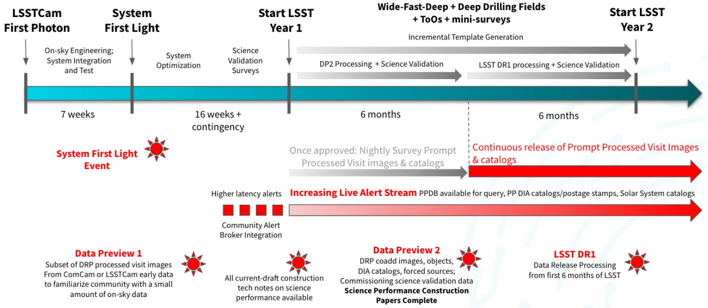

Figure 1: Detailed schedule of commissioning and early science activities relative to System

source: Rubin Observatory Plans for an Early Science Program Guy et al. (2025)

Early Science Program

Further information on the Early Science Program, and more, can be found at:

`https://rubinobservatory.org/for-scientists/resources`